EMBL's European Bioinformatics Institute (EMBL-EBI)

SureChEMBL2.0: A Patent Database with Chemical, Disease, and Protein Annotations

EMBL-EBI

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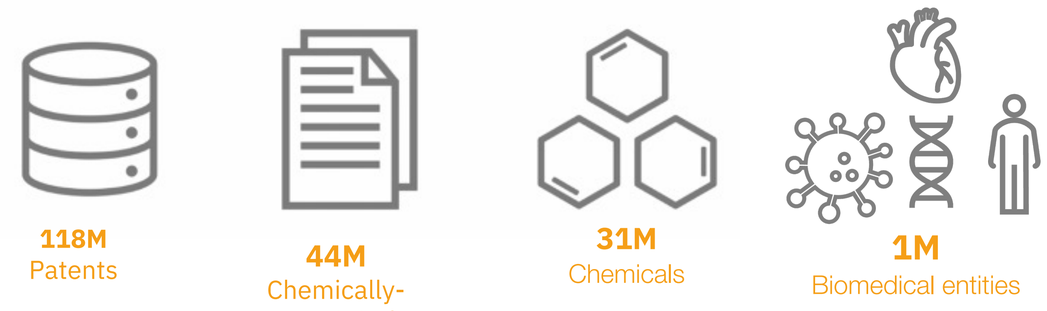
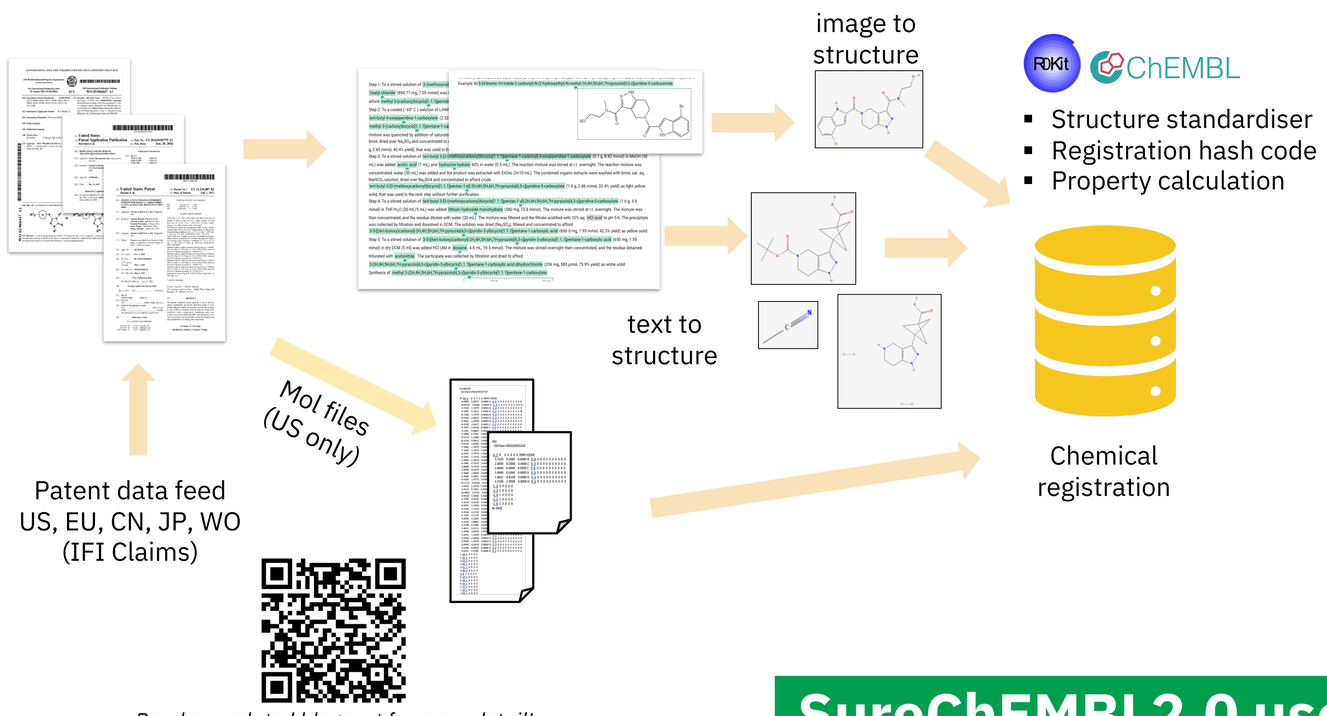
Patents describe and protect innovation, representing indirectly a rich source of information for drug development. First occurrences of compound and compound-target interaction often happen in the patent literature [1]. Despite the documents being freely accessible, data access remains challenging, and annotations are required for large-scale analysis. SureChEMBL [2] is a freely available resource that delivers extra value from patents, historically extracting chemical structures from text and images, but also more recently annotating the biomedical lexicon.
In 2025 we delivered SureChEMBL2.0, a major milestone offering a more robust pipeline for an improved data delivery. Here we give an overview of some recent changes, focusing in particular on the protein and disease annotation.



Read our related blogpost for more detail!

Compound identification pipeline

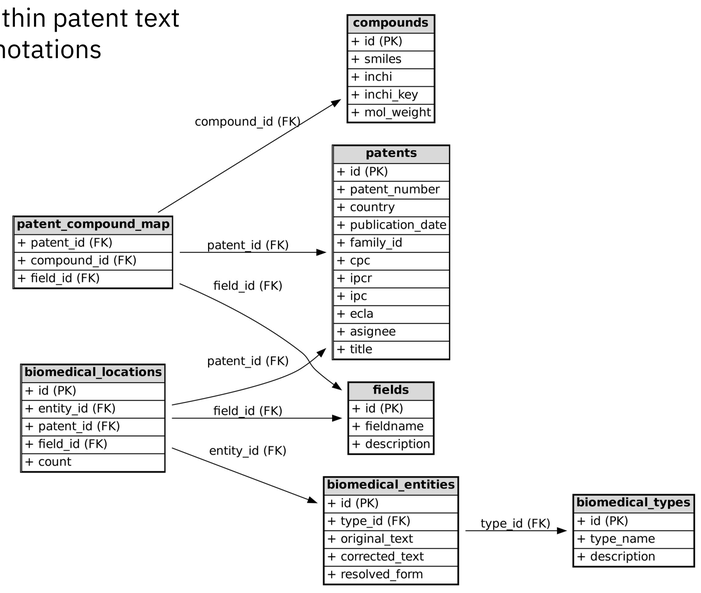


Read our related blogpost for more detail!

Bulk download

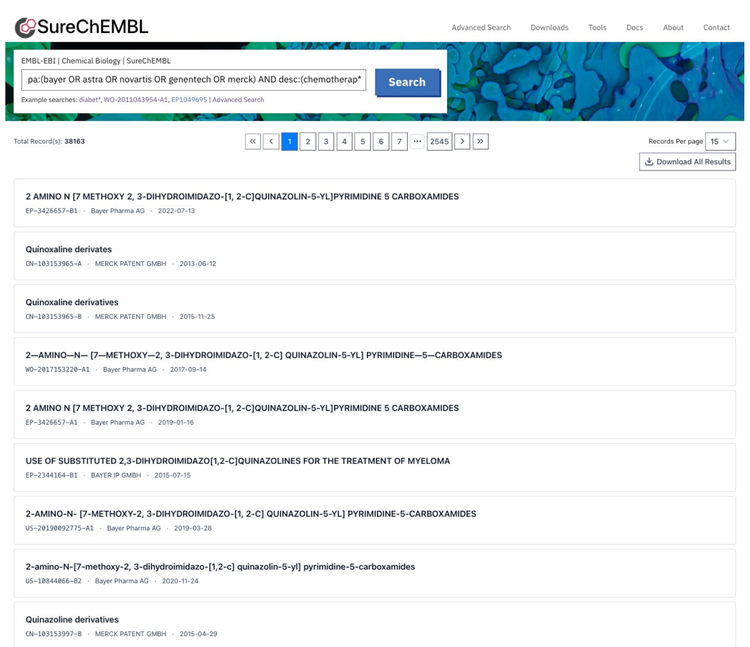
- Complete SureChEMBL snapshot every 2 weeks
- Highly compressed, yet easily queryable Parquet files
- Structured data includes:
  - Annotation locations within patent text
  - Protein and disease annotations
  - Chemical annotations
  - Patent metadata

Read our related blogpost for more detail!

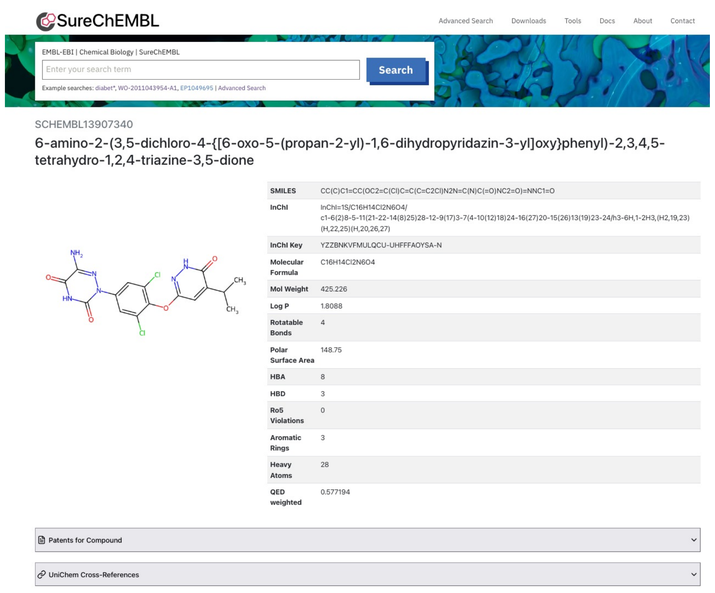


SureChEMBL2.0 user interface

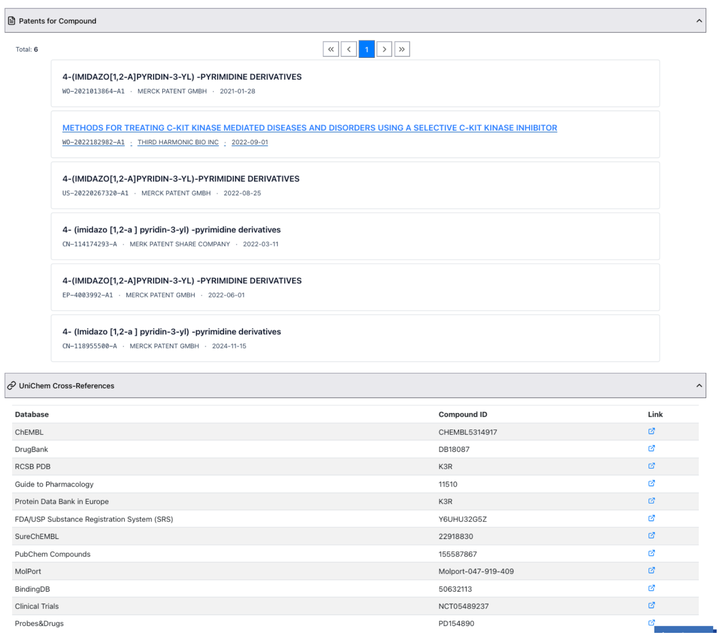
www.surechembl.org



Main page with search by keyword returning a paginated patent list matching the query



Compound details page with compound structure information

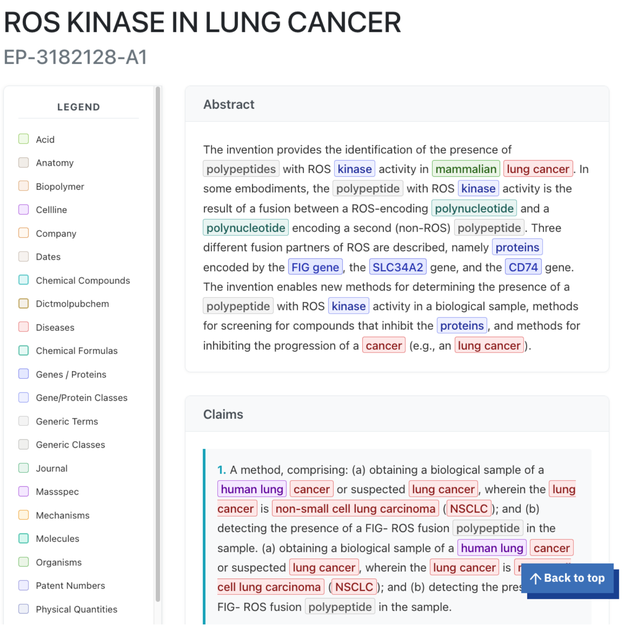


Compound details page with list of patents where the compound was found, and UniChem cross referencing

New text annotation system adds protein and disease annotation

- Leadmine (NextMove Software), text mining tool for the identification and annotation of chemicals, protein targets, genes, diseases, species, named reactions, company names, cell lines, etc. [3]
- Uses relevant dictionaries for identification and mapping
  - Gene/Protein: HGNC, Uniprot…
  - Disease: MeSH, Human Disease Ontology, NCBI MedGen…
  - Widely adopted across industry
- Deployment in SureChEMBL:
  - Chemical annotation already integrated
  - Protein and disease annotation being deployed

Read our related blogpost for more detail!



Patent annotation with Leadmine (early version not yet publicly available)

- Offers new answers to scientific questions:
  - Which diseases were first mentioned in 2025?

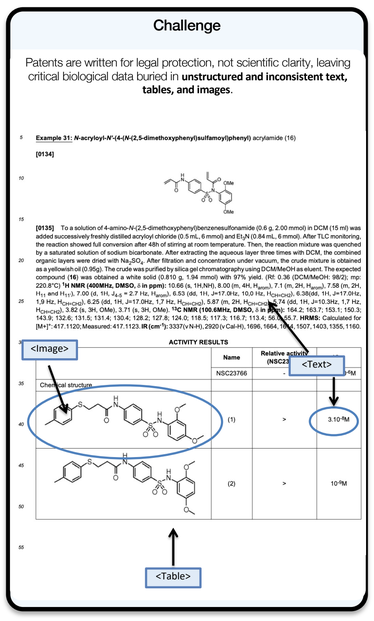
Text	Accessions	Patent count	Family count
Camptodactyly taurinuria	C537972	1	1
San Diego type skeletal dysplasia	C536670	1	1
St. Helena familial genu valgum	C537685	1	1
Velofaciosskeletal syndrome	C536536	1	1
cerebral intraventricular hemorrhage	D000074042	1	1

  - Which proteins were first mentioned in 2025?

Text	Accessions	Patent count	Family count
Sso0001	Q97TX9	4	2
Sso1391	Q97YD4	4	2
HcCYP	Q4G338	3	2
bik4	S0DZI6	3	3
Arlb protein	Q54UF1	2	2

  - But also
    - find patents combining given compound AND protein AND disease
    - Earliest mention of a compound
    - Etc.

- New on-going project for extracting bioactivity data from patent



[1] Senger, S. (2017). Assessment of the significance of patent-derived information for the early identification of compound–target interaction hypotheses. J. Cheminform. *9*, 26.
[2] Papadatos, G., Davies, M., Dedman, N., Chambers, J., Gaulton, A., Siddle, J., Koks, R., Irvine, S.A., Pettersson, J., Goncharoff, N., et al. (2016). SureChEMBL: a large-scale, chemically annotated patent document database. Nucleic Acids Res. *44*, D1220–D1228
[3] Lowe, D., Sayle, R. (2015). LeadMine: a grammar and dictionary driven approach to entity recognition. J. Cheminform. *7*, S1, S5.

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